Trainings

General Information

There were no changes made to the cylinder block when electronically actuated injectors were introduced into production.

There are two types of piston cooling nozzles used to cool each piston on the QSK45 and QSK60 engines:

- Inboard
- Outboard.

Both types are designed to work with gallery cooled pistons.

The piston cooling nozzles spray oil into a cast, located in the piston, for more effective cooling.

Single-Piece Piston

There have been two types of pistons in production for the QSK45 and QSK60 engines. Earlier engines were fitted with an articulated piston design, but were superseded in April, 2002 (Engine Serial Number First 33151184) with the single piece ferrous cast ductile iron pistons. The ferrous cast iron piston for the QSK45 and QSK60 engines are the same technology pistons used in the QST30 engine.

The ferrous cast ductile piston **must** be used in combination with a new hardened cylinder liner, piston ring set, and piston pin. The new cylinder liner is nitride hardened for improved robustness and supersedes the old cylinder liner. The new cylinder liner is dimensionally the same as the old cylinder liner. The ferrous cast ductile top piston ring is a cast iron ring with a wraparound hard particle chrome material and is narrower than the articulated top piston ring.

The piston pin is also unique to the ferrous cast ductile pistons. The ferrous cast ductile piston pin is similar in appearance to the K38 and K50 pistons, with square cut ends compared the articulated piston pin's concave ends.

When the new hardened cylinder liner is unused it is duller in appearance than the old liner because of the nitride hardening. However, the **only** certain method to determine the difference between the cylinder numbers is by the part number identification stamped into the cylinder liner.

Inspection of the top piston ring part number is also required to determine the correct piston ring part number. The top piston ring **must** be installed correctly, top side up, because the ring face is no longer symmetrical.

The ferrous cast ductile piston was designed to have the same weight as the articulated piston. As a result, both piston types can be intermixed in an engine in any number combination and regardless of engine bank.

The recommended service practice is to **not** reuse the articulated piston crown or the articulated piston skirt when the articulated piston is removed. Discard the articulated piston and cylinder liner. Install a ferrous cast ductile power cylinder, which includes the ferrous cast ductile piston, ferrous cast ductile piston ring set, ferrous cast ductile piston pin, and the hardened liner.

The QSK45 and QSK60 engines have one camshaft for the right bank and one camshaft for the left bank. The camshafts are different part numbers. The right and left camshaft rotate in different direction and have different cylinder spacing. The camshaft gears are different for the right and left banks of the engine. Camshaft gears for engines with electronically actuated injectors have slots on the backside of the gear so that the position sensor located in the accessory drive housing can read the camshaft position. Camshaft position is then provided to the electronic control module (ECM) to ensure proper timing of fueling. The camshafts and the camshaft gears are marked “R” or “RB” for the right bank and “L” or “LB” for the left bank of the engine. **Always** check the part number information to be sure the correct parts are installed.

Engines with Mechanically Actuated Injectors.

The injection timing is adjusted with the use of different camshaft keys. The selection of a key changes the position of the camshaft lobes in relation to the timing mark on the camshaft gear. The camshaft gear **must** be removed to change the injection timing.

The camshaft end clearance is determined by the clearance between the camshaft and the thrust bearing plate. The camshaft gear **must** be removed to adjust the camshaft end clearance.

The camshaft does **not** need to be removed to remove the camshaft gear.

Camshafts that are damaged or worn on the injector or the valve lobes **must** be replaced. Cummins Inc. does **not** recommend grinding of camshaft lobes or rollers.

If the engine is found to be magnetized, the engine will need to be disassembled and examined for:

- Magnetism of ferrous components
- Debris through the lubrication system
- Damage to bearings and bushings. This includes: crankshaft thrust bearings, main bearings, connecting rod bearings, and camshaft bushings.
- Component wear or damage.

Acceptable Magnetism Levels	
Injectors and Ferrous Injector Components	**All Other Ferrous Engine Components**
5 or less gauss units	15 or less gauss units

Measure each ferrous component with a gauss meter and record the results. If the magnetism is out of specification, the engine **must** be treated in general as if:

- Debris (fine particles) have been traveling throughout the lubrication system, resulting to wear and damage to the components.

Refer to Section DS for engine disassembly procedures and Section AS for engine assembly procedures.

Components with measured gauss units greater than specification **must** be demagnetized or replaced. A facility capable of magnetic testing (Magnaflux) of the engine components is capable of demagnetizing (degaussing) components.

Check components in the lubrication system closely and thoroughly clean the oil galleries.

Replace all main bearings, thrust bearings, connecting rod bearings, and camshaft bushings.

Note : Do not attempt to demagnetize sensors, engine control modules, or actuators.

There are two methods to demagnetize allowed components:

- Passing the part through and alternating coil (50 or 60 cycles per second)
- Passing a reversing, 30 point step-down current through the part.

The alternating current coil is suggested for smaller parts.

For larger mass parts, the reversing 30 point step-down is suggested.

Parts with acceptable levels of magnetization **must** be cleaned and inspected for reuse.

It is recommended to have the alternating current coil just large enough for the parts to pass through. A small part passed through a large coil will **not** be demagnetized as well as if it were passed through a smaller coil.

The coil **must** be located so the longest axis of the part is perpendicular to the coil when passing through. Parts **must** pass through and a minimum of 457.2 to 609.6 mm [18 to 24 in] beyond the coil for the most effective demagnetization.

Do **not** attempt to demagnetize small parts by loading them into a basket and passing the basket through a coil. Do **not** attempt to demagnetize a whole engine assembly.

Direct current demagnetization can be accomplished by using the magnetizing unit. Clamp the part between the head and the tail stock. Activate the demagnetization controls, and the reversing, step-down current passes through the part. Check all parts with the gauss meter.

When tiny pits occur in clearly defined patterns, or surfaces are fluted, electric current can be the problem. The patterns will vary with metals, sources, and movement.

For insert bearings, pitting is a chief indication. In antifriction bearings, such as ball bearings, fluted surfaces or wavy lines of pitting in patterns differing with rotation, vibration, and current.

- Electrically actuated components (such as a clutch).
- Static current from belts or other moving parts.
- Grounding of the electrical system through the crankshaft when some component such as the generator or engine block has **not** been grounded properly. A six volt system improperly grounded can cause damage.

Engine that have experienced dust outs because of intake air filtration system resulting in cylinder liner and piston ring wear. These engines **must** be treated as if the lubricating oil system has been contaminated with debris. The engine **must** be disassembled and cleaned appropriately. The camshafts and cam followers (valves and injector) **must** be cleaned and examined for wear.